

extension to the PBX. Outside the office, these can be normal GSM phones; inside the office, they turn into dual phones, which can also act as extensions.

Roaming extension: Doctors roaming around hospitals, work-site supervisors, hotel staff, hotel room guests, etc., are rarely found at their desktops or in their rooms. With Wi-Fi technology, they are always logged into the IP PBX and are, therefore, reachable.

Cost savings: The user is always logged into the PBX and therefore can route all the calls through the PBX, which could be cheaper, especially when the volume of calls is high.

FX or 'foreign exchange'

FXO or FXS are used to denote various connections in analogue technology (or POTS – Plain Old Telephone System), which is the most common SOHO (Small Office Home Office) technology prevalent in India today. FX stands for Foreign eXchange. FXO (Foreign eXchange Office) is the line which connects to the local telecom switch and FXS (Foreign eXchange Subscriber) connects to the phones or other end user devices like fax machines.

VoIP FXS gateway or FXS card

Analogue telephones are the cheapest user equipment available, with prices starting as low as ₹ 300 compared to a VoIP telephone starting at ₹ 3000. An FXS gateway/card connects an analogue phone to the Asterisk server. Most of the interface hardware comes in two forms, i.e., PCI/PCIE cards, which can be inserted into the server or external standalone gateways connected to the server over IP. These devices can have multiple ports—the common ones are FXS cards with up to 16 ports or gateways with up to 48 ports. For a larger number of ports, it is advisable to use multiple gateways.

VoIP FXO gateways or cards

Analogue trunks are the most common trunk connection available for PBX equipment to connect to the outside world. With the help of a VoIP FXO gateway or card, one can connect the Asterisk equipment to the PSTN world. Their dimensions or capabilities are given as 1-, 2-, 4-, 16-port, etc.

VoIP PRI gateway or card

ISDN (Integrated Services Digital Network), which is purely a digital network, is the next generation of telecommunication networks, a step up from analogue networks. PRI (Primary Rate Interface) and BRI (Basic Rate Interface) are the two common connection types available. PRI is a 4-wire interface to the customer, and can carry 30 B channels (voice channels) and one D channel (control channel) in an E1 configuration, the most common in the Indian scenario. There is also a T1 configuration with 24 B channels and one D channel, which is more prevalent in the US. The BRI with two B-channels and one D-channel is meant for SOHO use. In India, generally

only PRIs are used in the corporate scenario, and BRI is non-existent, for all practical purposes.

The PRI card can be inserted into the server PCI or PCIE slot, and can be configured as NT (Network Terminator or trunk side equipment) or TE (Terminal Equipment or subscriber side equipment). A PRI connection can have one pilot number and multiple subscriber numbers or DIDs (Direct Inward Dialling), typically 100.

VoIP GSM gateway or card

The mobile network, specially the GSM mobile network, is the most popular and widespread network nowadays. The GSM gateway or card makes it possible to connect Asterisk directly to this network. This equipment resembles a mobile phone, where a SIM card can be inserted, and comes in 1, 2 or 4 ports. The GSM network can also be used as a backup for the POTS or ISDN lines.

For large density connectivity, VoIP GSM gateways are suitable, which come in 8, 16 and 32 ports.

Echo cancellation hardware

An echo is the phenomenon by which sound is repeated, which in turn affects the clarity of the sound. Asterisk has built-in software routines to combat this problem. In cases where the software routines are not effective, DSP-based hardware is available for analogue as well as PRI hardware. Most of them are 'piggy back' modules, which have to be plugged on to the existing boards.

Voice logger

Conversations can be logged within Asterisk without any additional device. A command in dialplan is enough to activate the voice logging feature in Asterisk for calls passing through the Asterisk equipment.

Asterisk can also operate as a voice logger system for other systems in an active or passive mode. In both cases, the line to be tapped has to loop through the voice logger system. In the case of active voice loggers, interface cards with double ports of the interfaces to be tapped are required, i.e., to tap one PRI, we need a 2-port PRI card. The line to be tapped is connected to the voice logger system and the destination system also. When a call comes, the connection is established to the voice logger system, and then again a back-to-back connection is generated to the destination system. The calls are recorded in a similar manner in an Asterisk system using dialplan.

In case of passive voice loggers, a passive interface card with double the number of interfaces to be tapped is connected. The main difference is that there is no connection established to the voice logger system. The voice logger system passively taps the required information and stores it.

Redundancy hardware

Redundancy hardware can be created easily by having a mirror copy of the existing server. In case of a crisis, when